

```

# This Python 3 environment comes with many helpful analytics
libraries installed
# It is defined by the kaggle/python Docker image:
https://github.com/kaggle/docker-python
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/"
directory
# For example, running this (by clicking run or pressing Shift+Enter)
will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/)
that gets preserved as output when you create a version using "Save &
Run All"
# You can also write temporary files to /kaggle/temp/, but they won't
be saved outside of the current session

import pandas as pd

df = pd.read_csv('/kaggle/input/titanic-datasett/train.csv')
df.head()

```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	SibSp	\	Name	Sex	Age
0			Braund, Mr. Owen Harris	male	22.0
1					
1			Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0
1					
2			Heikkinen, Miss. Laina	female	26.0
0					
3			Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0
1					
4			Allen, Mr. William Henry	male	35.0
0					

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

```
df.shape
df.info()
df.describe()
df.isnull().sum()
df['Survived'].value_counts()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

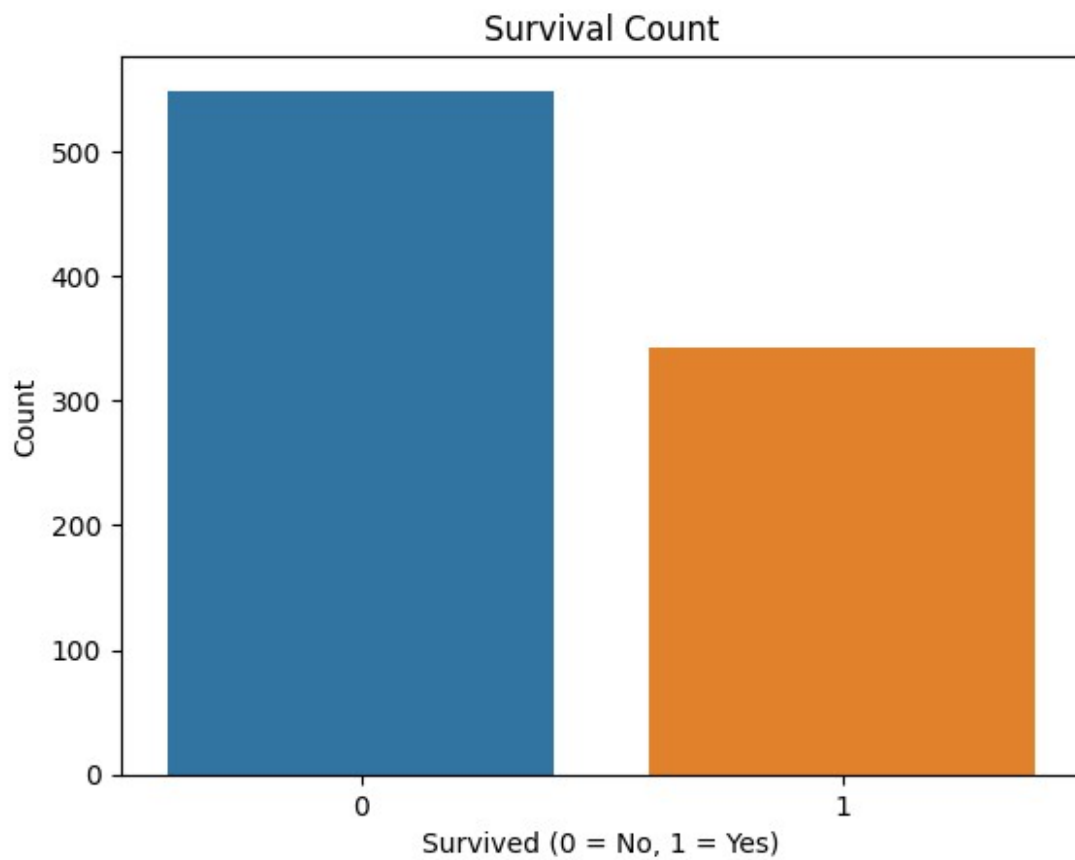
```
Survived
0    549
1    342
Name: count, dtype: int64
```

```
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
df = df.drop(columns=['Cabin'])

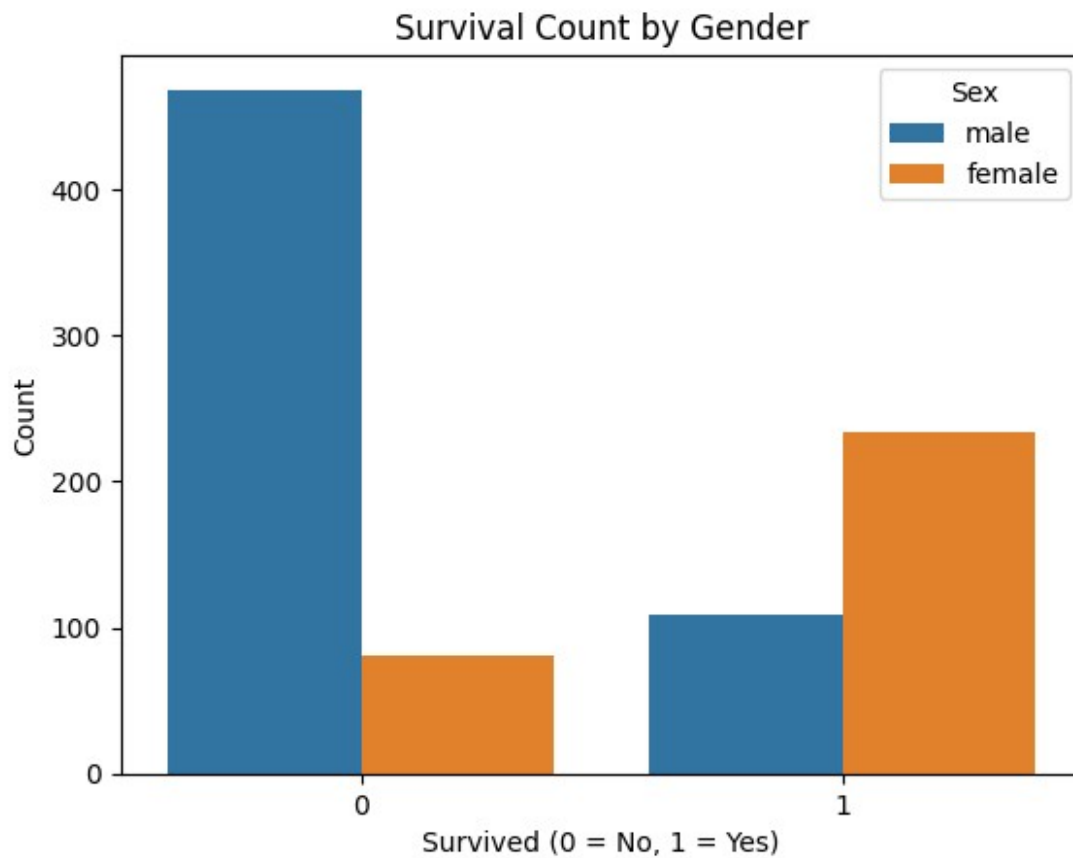
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
```

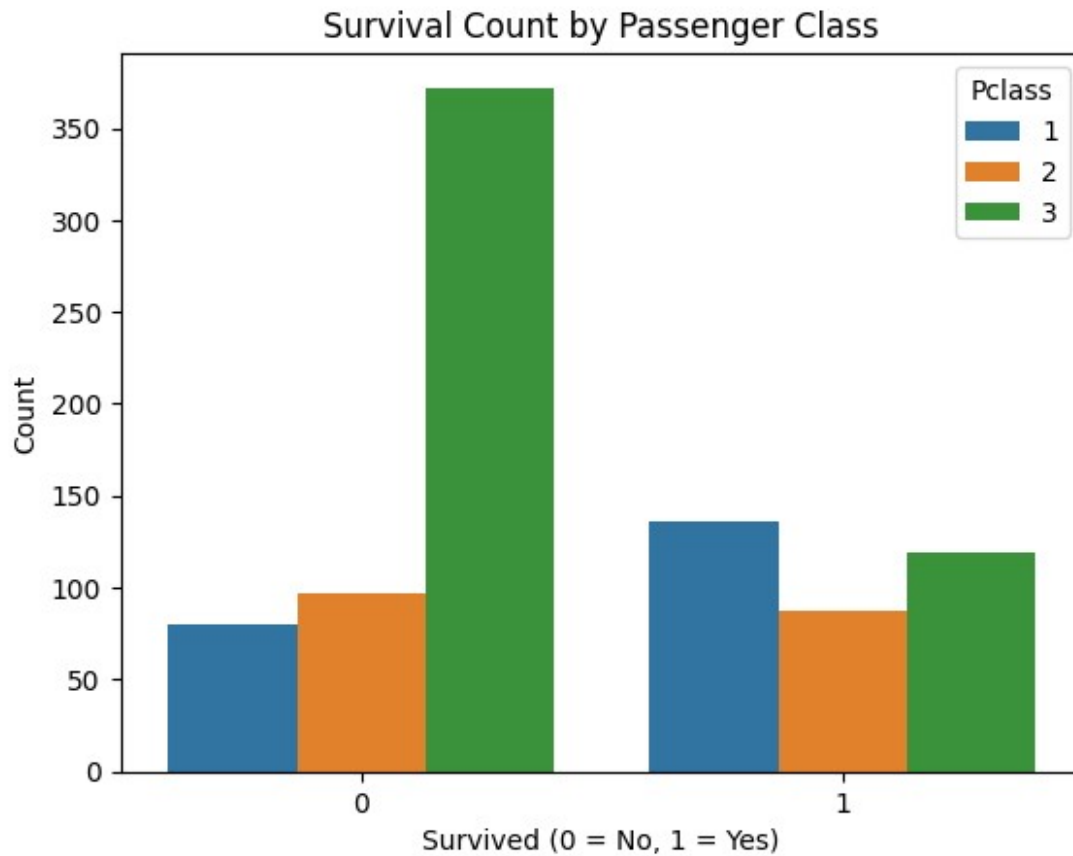
```
plt.ylabel("Count")  
plt.show()
```



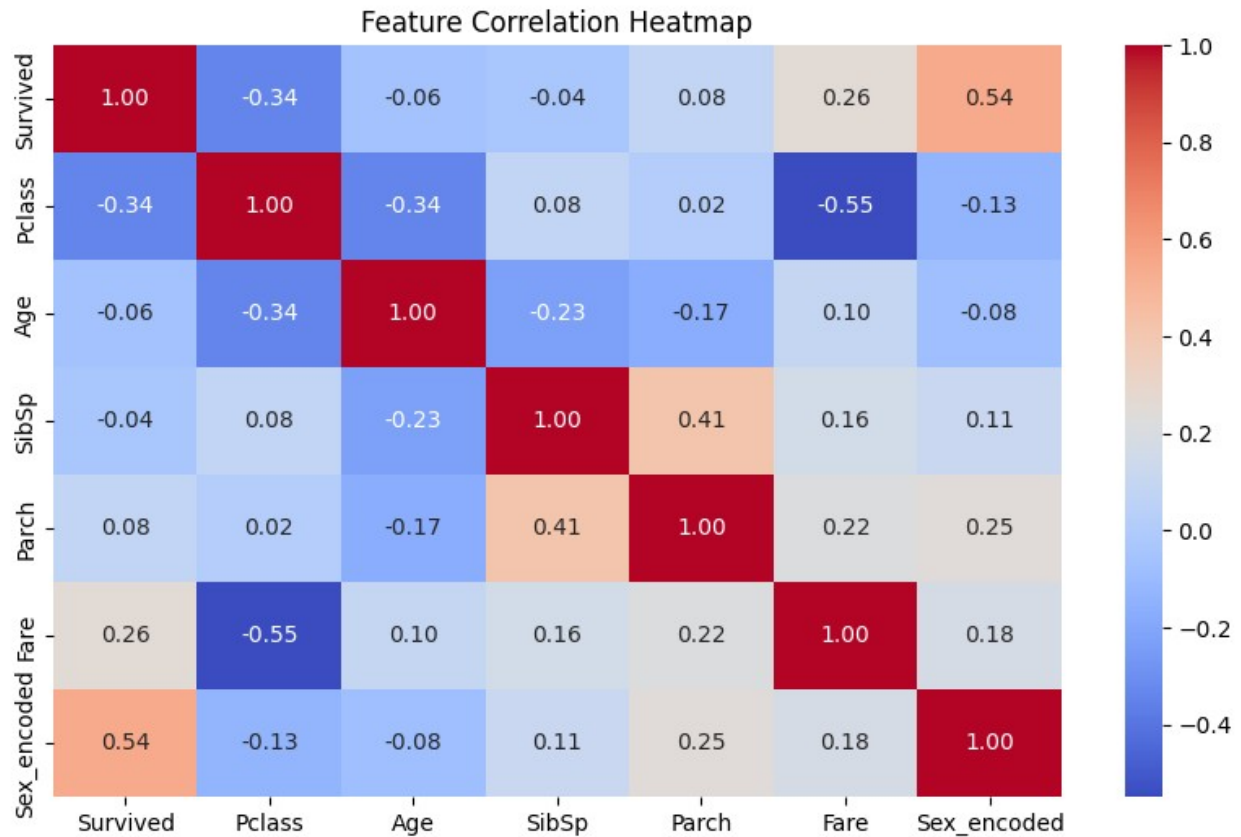
```
sns.countplot(x='Survived', hue='Sex', data=df)  
plt.title("Survival Count by Gender")  
plt.xlabel("Survived (0 = No, 1 = Yes)")  
plt.ylabel("Count")  
plt.show()
```



```
sns.countplot(x='Survived', hue='Pclass', data=df)
plt.title("Survival Count by Passenger Class")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.show()
```



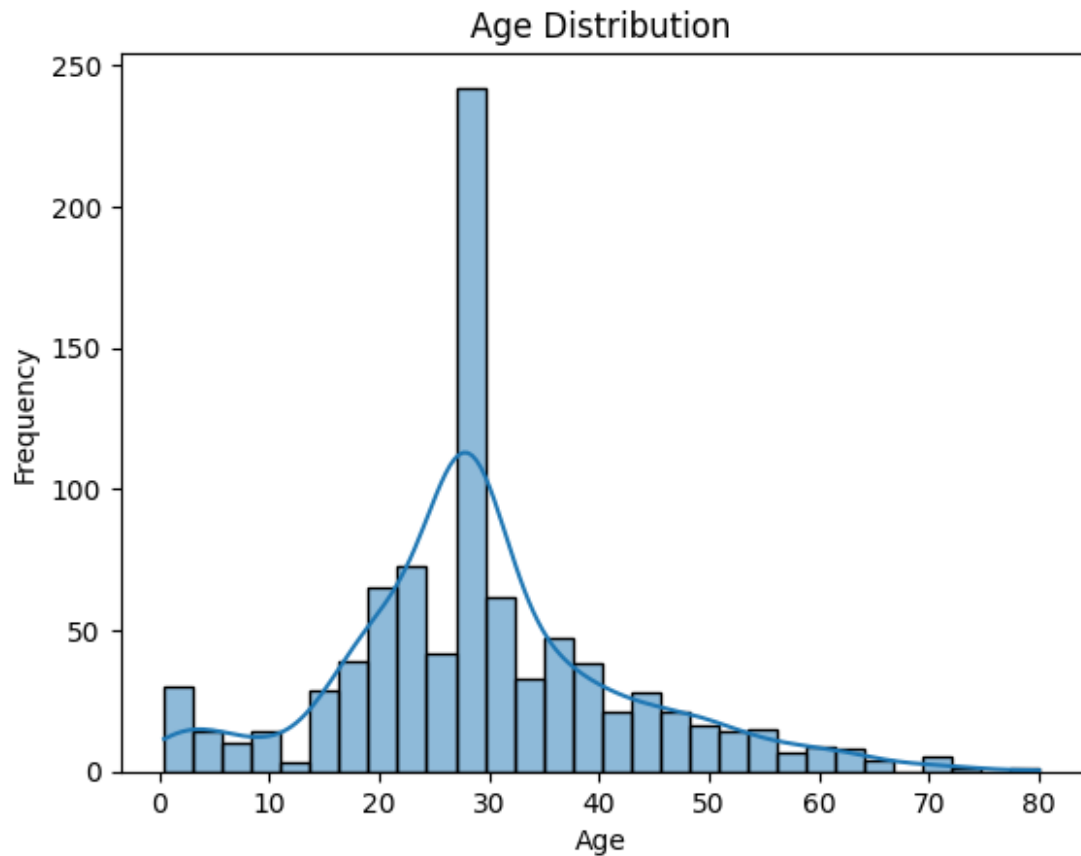
```
import numpy as np
df['Sex_encoded'] = df['Sex'].map({'male': 0, 'female': 1})
corr_matrix = df[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch',
                  'Fare', 'Sex_encoded']].corr()
plt.figure(figsize=(10, 6))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Feature Correlation Heatmap")
plt.show()
```



```
sns.histplot(df['Age'].dropna(), kde=True, bins=30)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```

/usr/local/lib/python3.11/dist-packages/seaborn/_oldcore.py:1119:
FutureWarning: use_inf_as_na option is deprecated and will be removed
in a future version. Convert inf values to NaN before operating
instead.

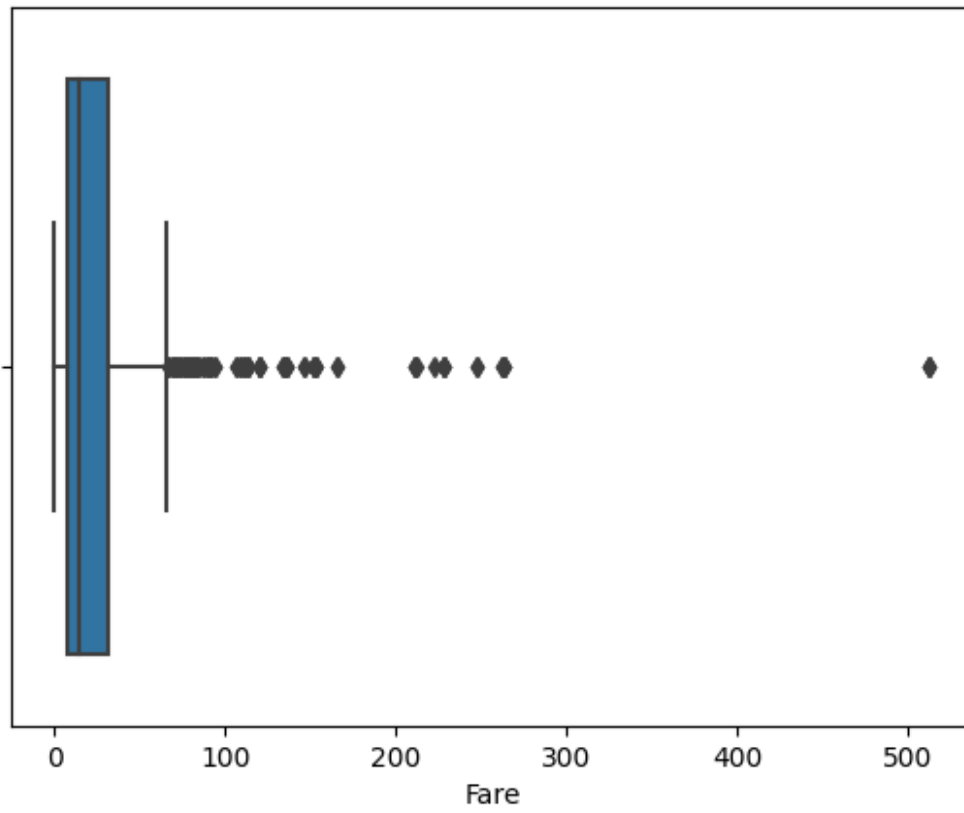
```
with pd.option_context('mode.use_inf_as_na', True):
```

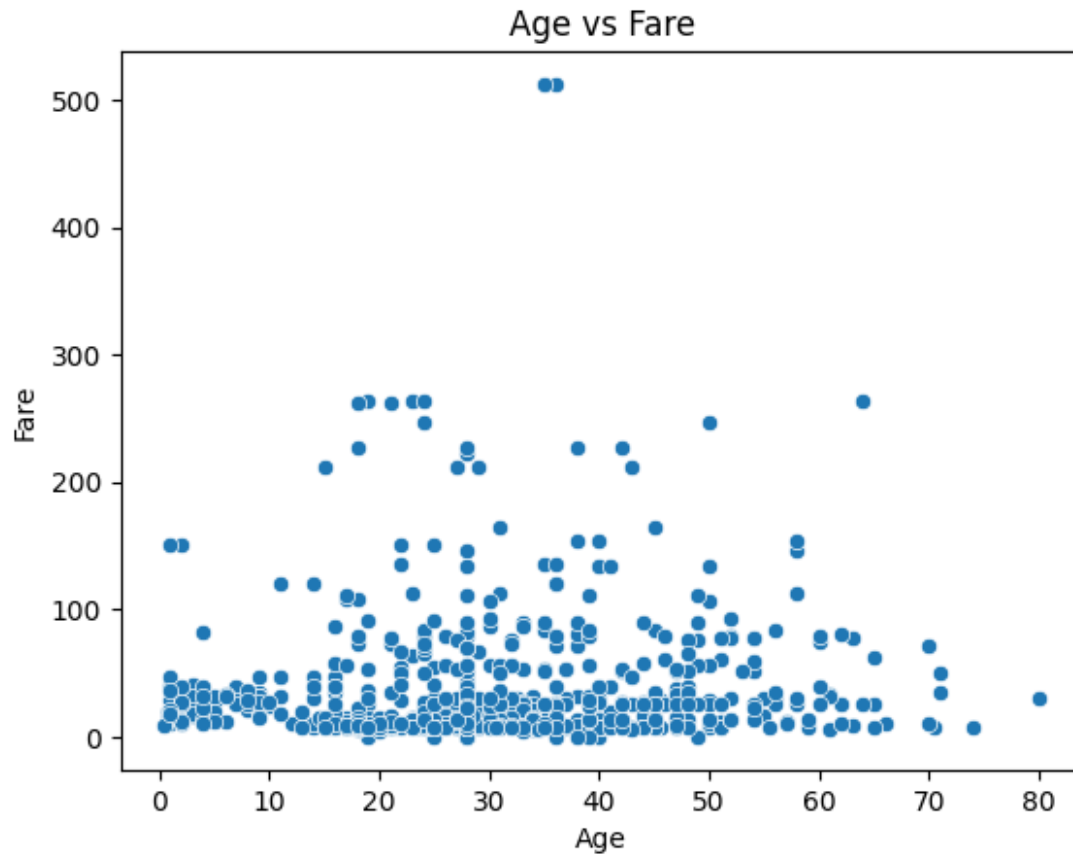


```
sns.boxplot(x='Fare', data=df)
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.show()

sns.scatterplot(x='Age', y='Fare', data=df)
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

Fare Distribution

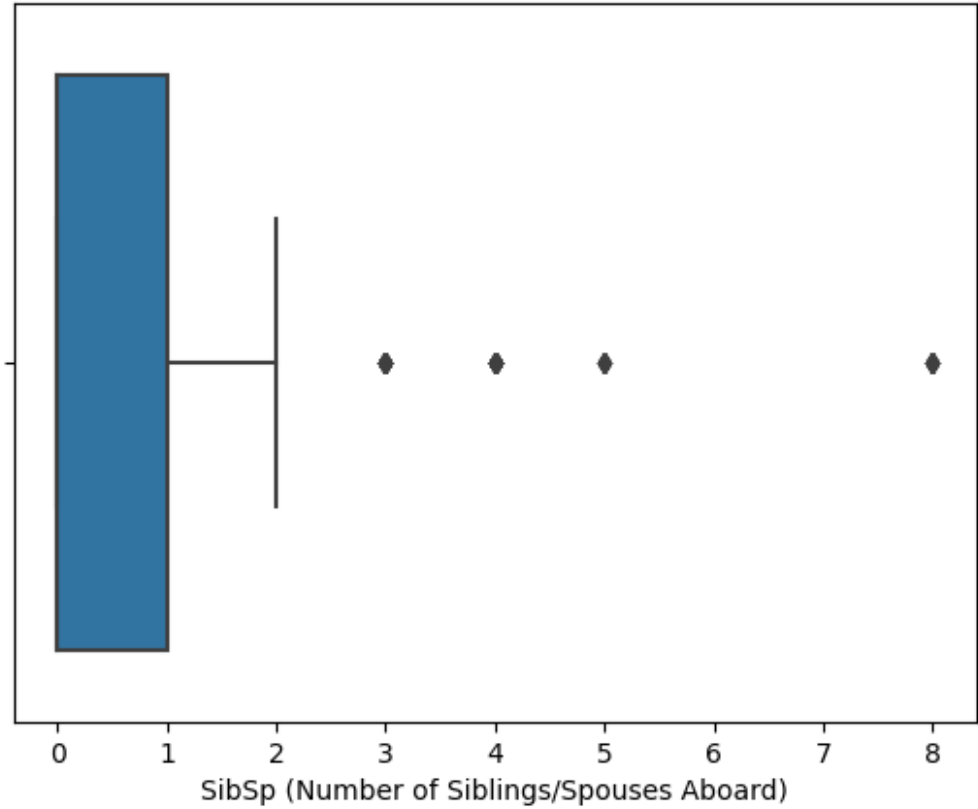


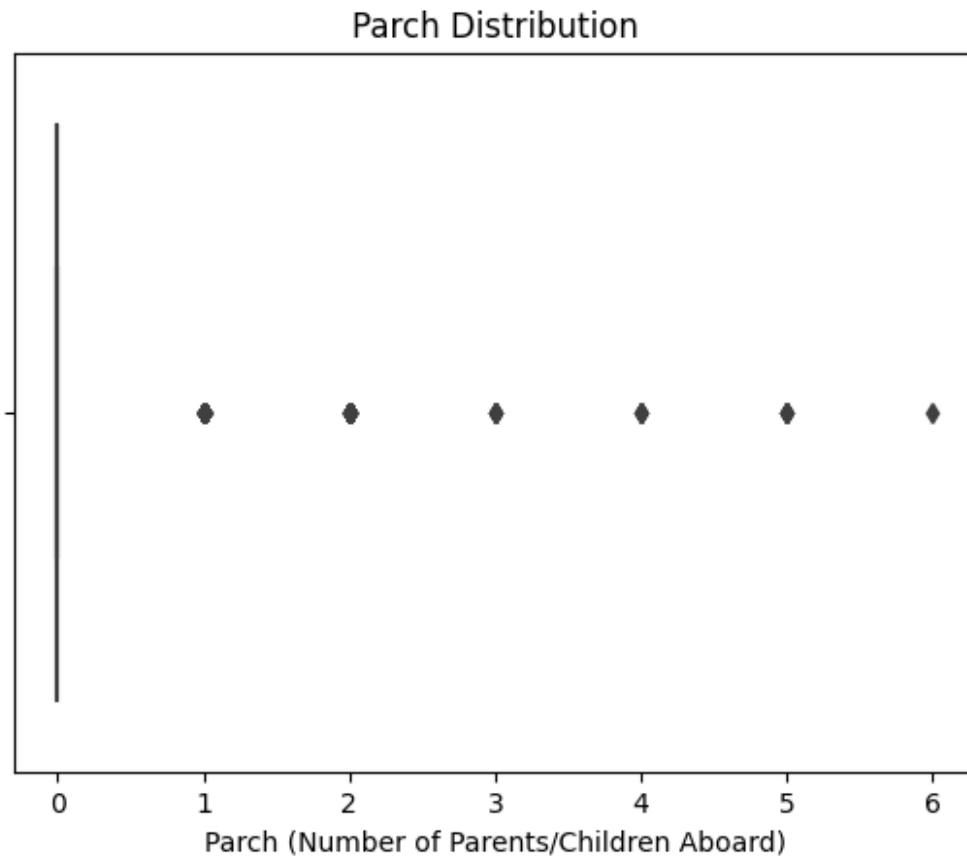


```
sns.boxplot(x='SibSp', data=df)
plt.title("SibSp Distribution")
plt.xlabel("SibSp (Number of Siblings/Spouses Aboard)")
plt.show()

sns.boxplot(x='Parch', data=df)
plt.title("Parch Distribution")
plt.xlabel("Parch (Number of Parents/Children Aboard)")
plt.show()
```

SibSp Distribution





```
corr_matrix = df[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch',  
'Fare', 'Sex_encoded']].corr()  
plt.figure(figsize=(10, 6))  
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")  
plt.title("Feature Correlation Heatmap")  
plt.show()
```

